

MEC V1.0 - Modular Embedded Controller

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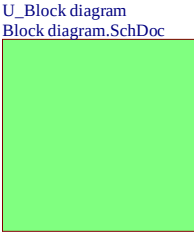
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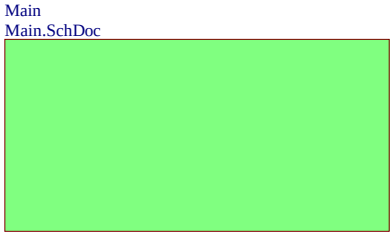
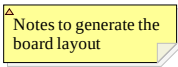
Sheet 10: Microcontroller (MCU)

Uniformar librería (BOM)
Uniformar librería Schematics
Uniformar librería Footprints según IPC



Legend

- General comment such function title, configurations, datasheets, ...
- Text to be added to silkscreen
- Design references
- Warning text
- DNP Do-not-place



Title:	Project overview.SchDoc		
Project:	MEC.PrjPcb		
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Draw by:	Ing. Lucas Kirschner		



1

2

3

4

A

A

B

B

C

C

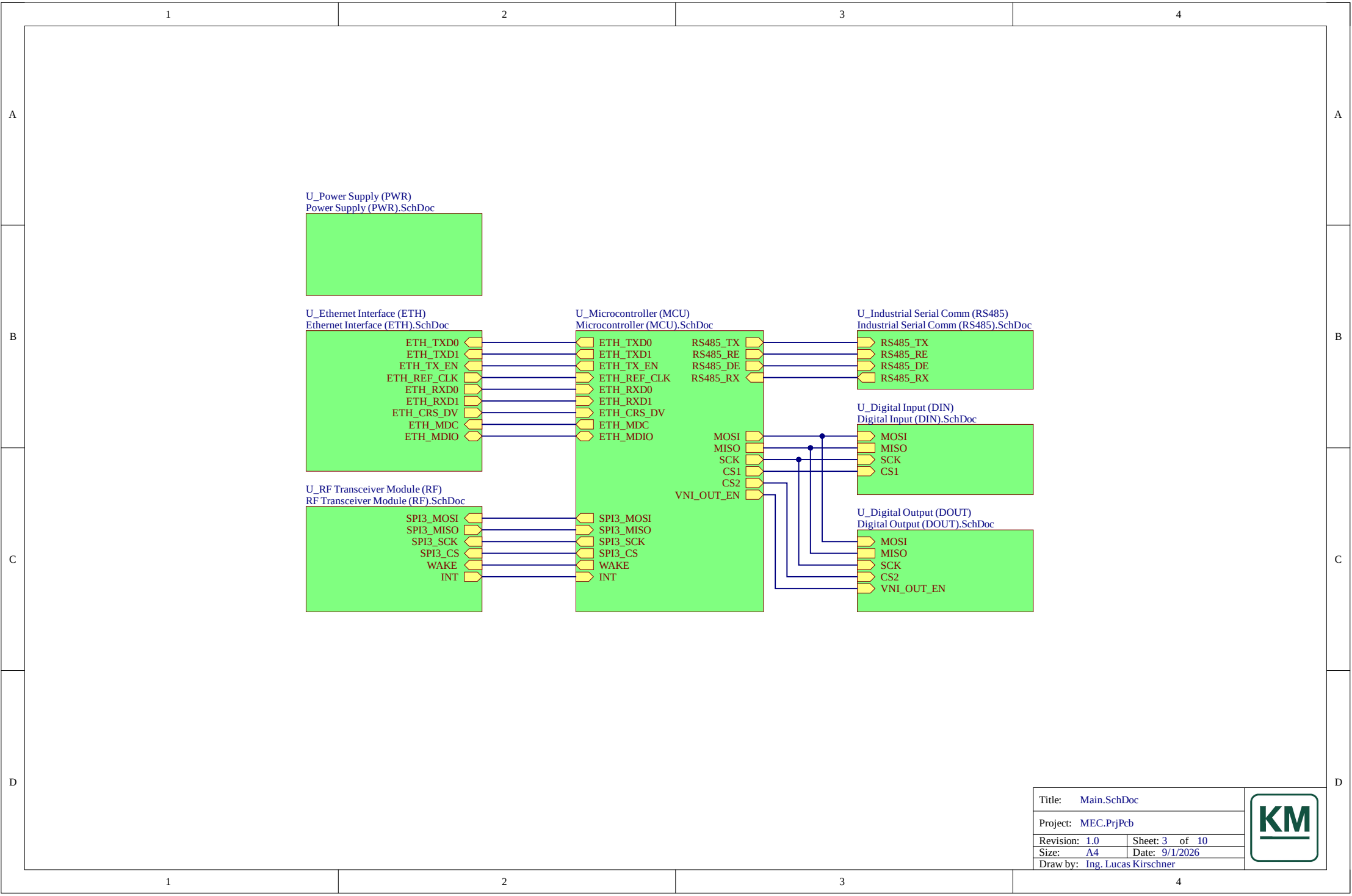
D

D

Reservado diagrama de bloques

Title: Block diagram.SchDoc	
Project: MEC.PrjPcb	
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Draw by: Ing. Lucas Kirschner	





[illegible]

3V3 linear regulator

The diagram shows a 3V3 linear regulator circuit. The integrated circuit (U6, LD1117DT33CTR) is represented by a yellow box with pins 2, 3, and GND. Pin 2 (VIN) is connected to a 5V input and has a 100nF capacitor (C57) to ground. Pin 3 (VOUT) is connected to a 3V3 output and has a 10uF capacitor (C58) to ground. The output is also labeled TP3. The text '3.3V / 0,4A' is displayed above the output. The IC is labeled 'U6 LD1117DT33CTR'.

GND NET Tie

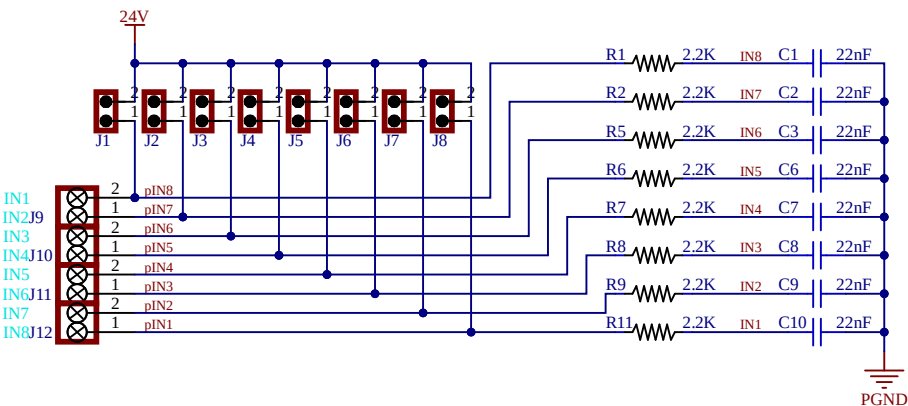
The diagram illustrates a GND NET Tie using a diode. Two ground planes, PGND and SGND, are shown as red horizontal lines. A diode, labeled D25 and SSA210, is connected between them. The anode of the diode is connected to PGND at a test point labeled TP4. The cathode of the diode is connected to SGND at a test point labeled TP5. The diode symbol is shown with the triangle pointing towards the PGND side.

References

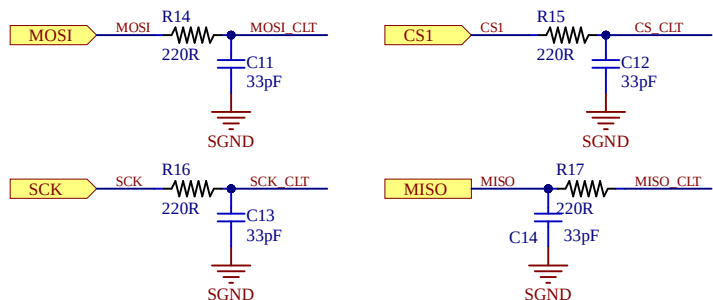
- 1) El conector de 24 V puede funcionar como entrada o salida, según la presencia o ausencia del módulo HLK-50M24C en el circuito. No aplicar 24 V externos si el módulo HLK-50M24C está instalado.



Digital input connector



SPI signals

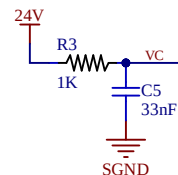


References

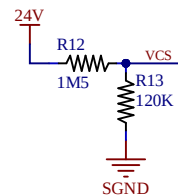
See <https://www.st.com/en/protections-and-emi-filters/sclt3-8bt8>
 See <https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp031v1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp007v1>

High speed digital input current limiter

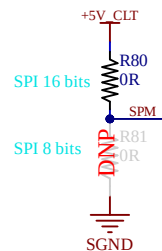
24 V sensor power supply



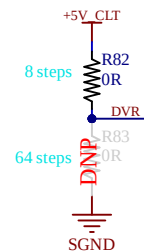
24 V sensor power supply sensing input



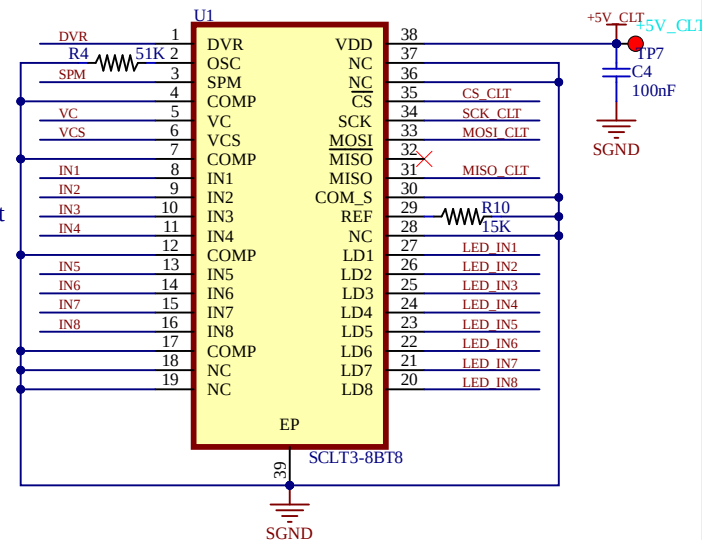
SPI 8/16-bit selection



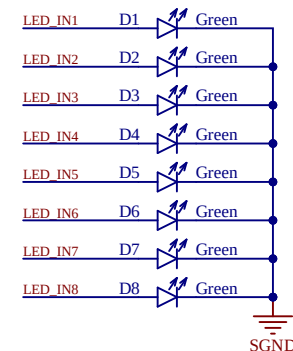
Divider ratio selector of the digital input filters



VDD max power supply: 12mA



LED input driver



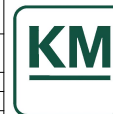
Title: Digital Input (DIN).SchDoc

Project: MEC.PrjPcb

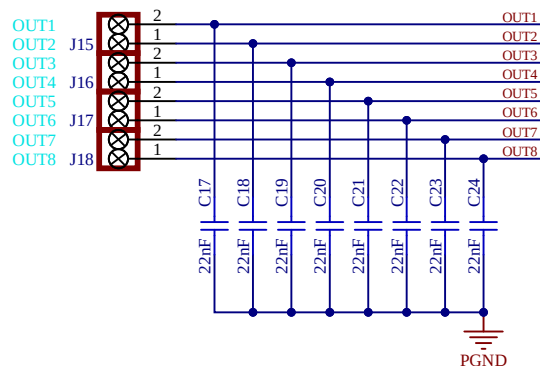
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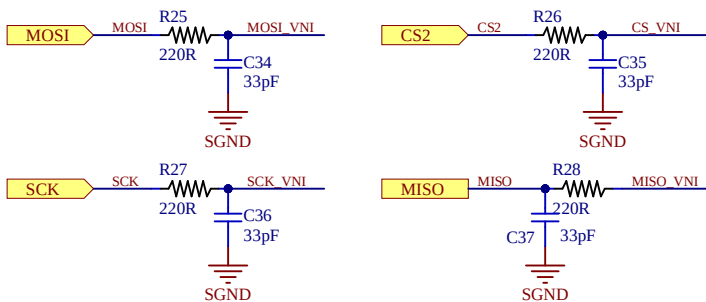
Draw by: Ing. Lucas Kirschner



Digital output connector



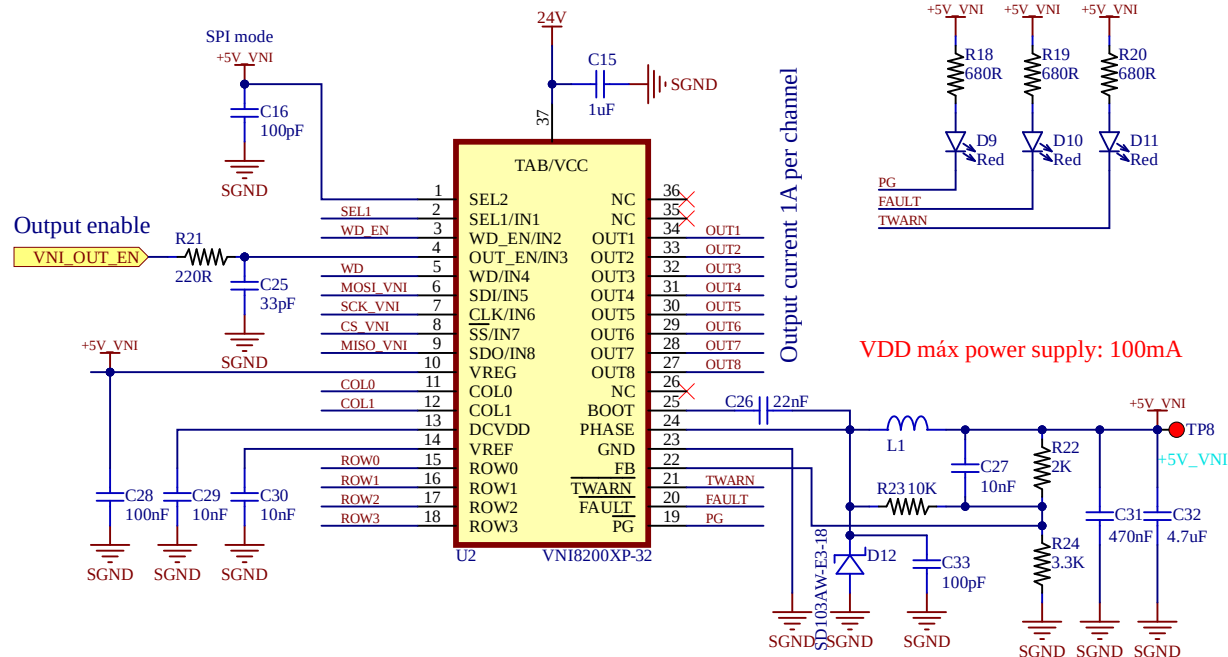
SPI signals



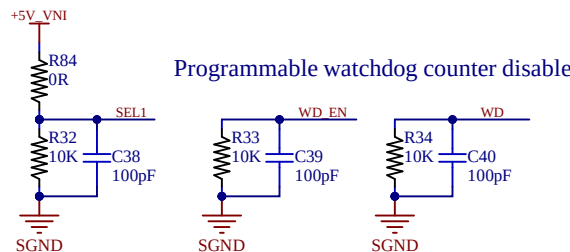
References

See <https://www.st.com/en/power-management/vni8200xp-32>
 See <https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp032v1>
 See <https://www.st.com/en/evaluation-tools/steval-ifp022v1>

Octal high-side smart power interface

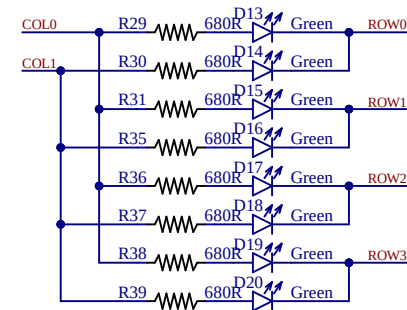


SPI 8/16-bit selection



Programmable watchdog counter disable

LED output driver



Title: Digital Output (DOUT).SchDoc

Project: MEC.PrjPcb

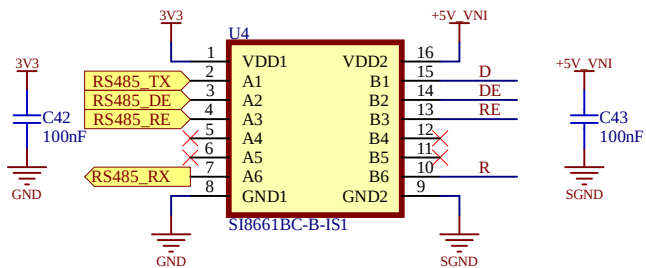
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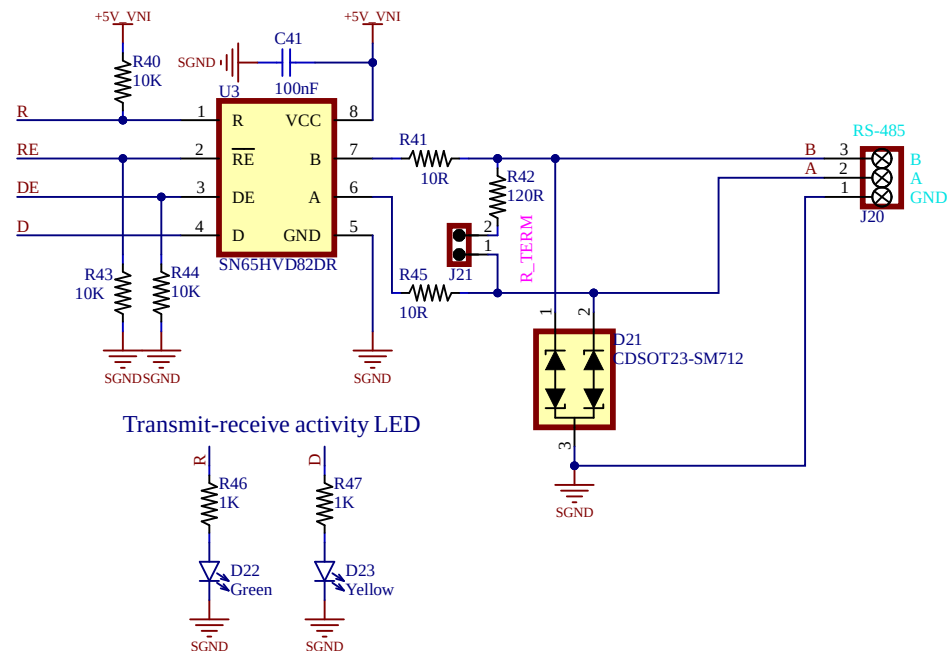
Draw by: Ing. Lucas Kirschner



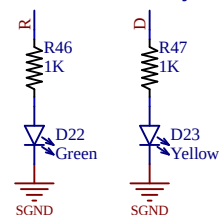
Digital isolator



RS485 interface



Transmit-receive activity LED



A: Driver output or receiver input (complementary to B)
 B: Driver output or receiver input (complementary to A)
 D: Driver data input
 DE: Driver enable, active high
 R: Receive data output
 RE: Receiver enable, active low

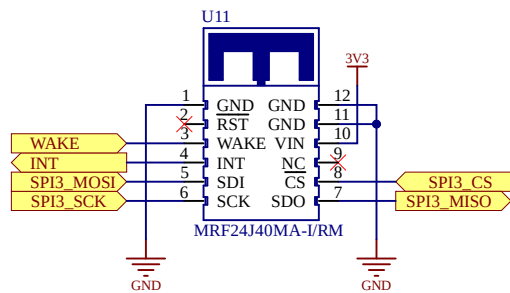
References

See <https://www.ti.com/product/SN65HVD82>
 See <https://www.st.com/en/evaluation-tools/x-nucleo-plc01a1>
 See <https://www.ti.com/tool/RS485-HF-DPLX-EVM>
 See <https://www.ti.com/lit/an/slla070d/slla070d.pdf>

Title: Industrial Serial Comm (RS485).SchDoc	
Project: MEC.PrjPcb	
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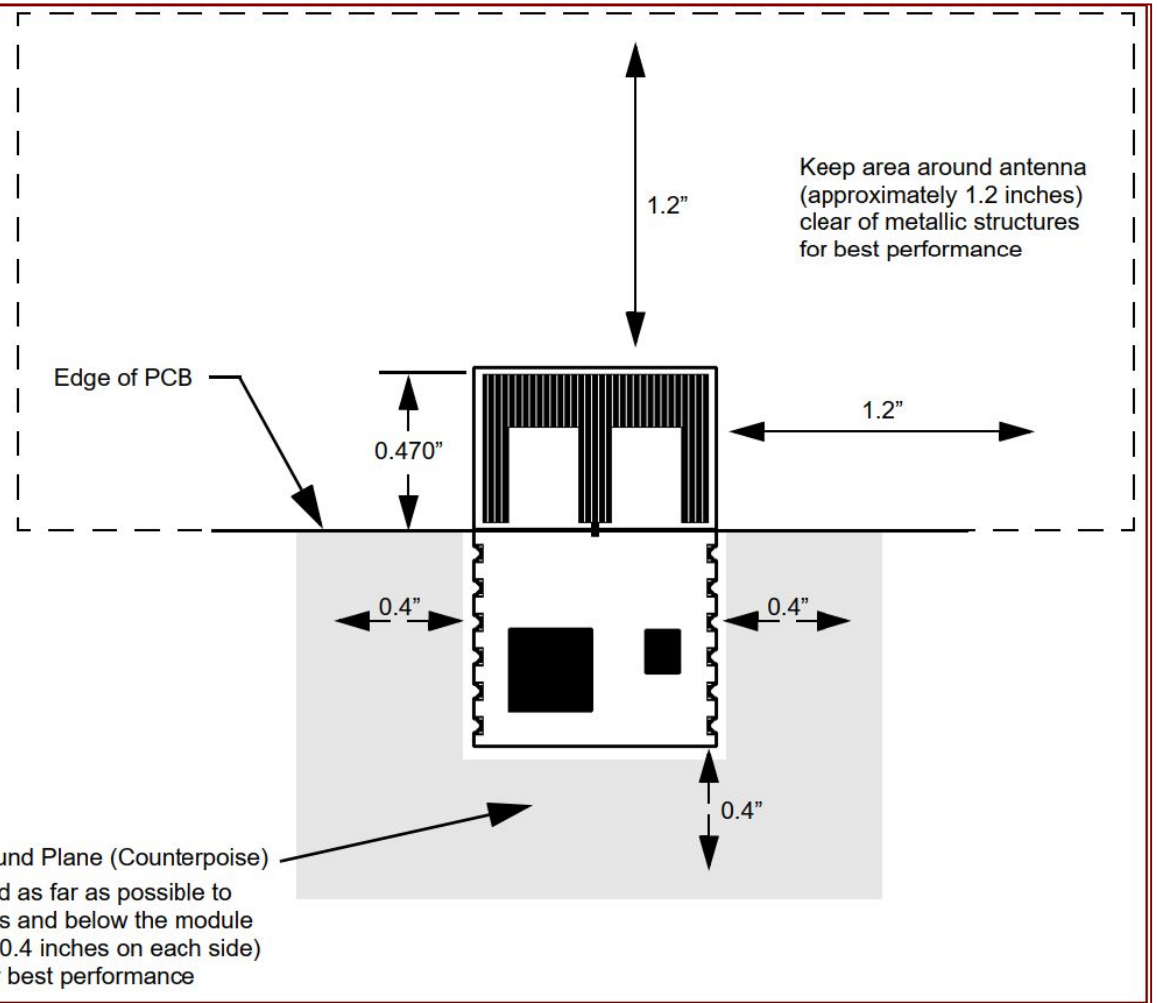


RF Transceiver Module



References

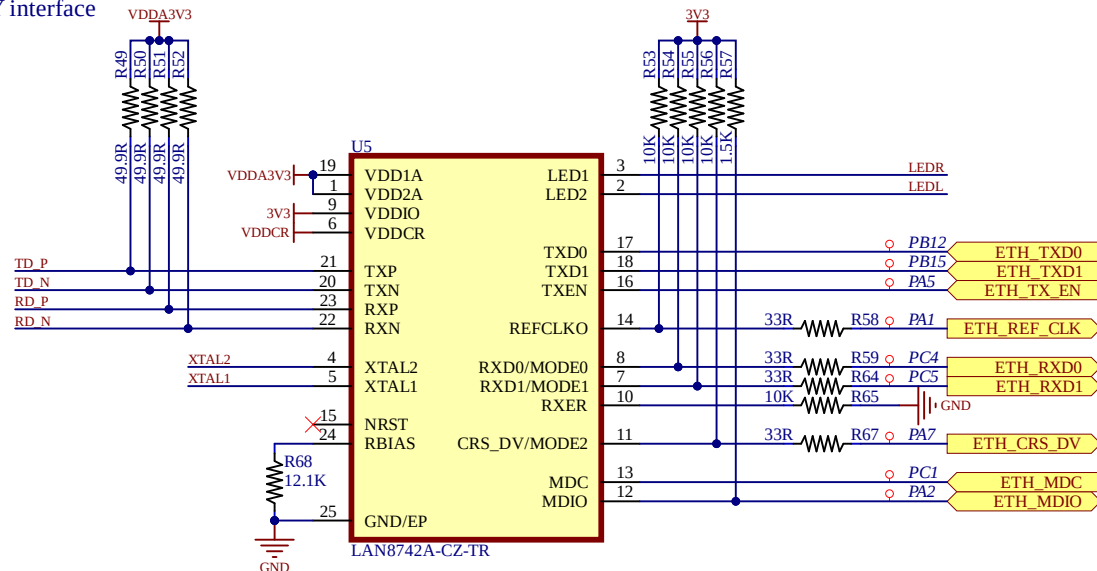
See <https://www.microchip.com/en-us/product/MRF24J40MA>
Pull-Up interna en RST.



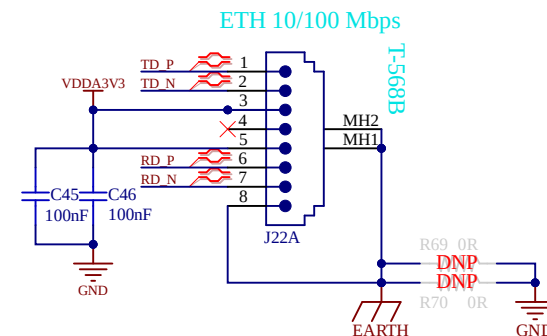
Title: RF Transceiver Module (RF).SchDoc	
Project: MEC.PrjPcb	
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Draw by: Ing. Lucas Kirschner	



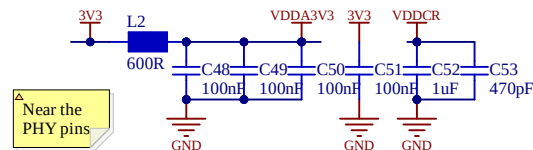
PHY interface



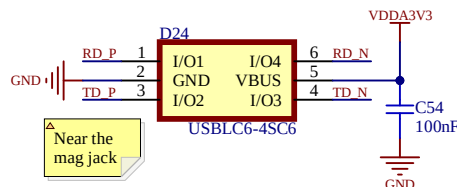
Ethernet transformer & connector



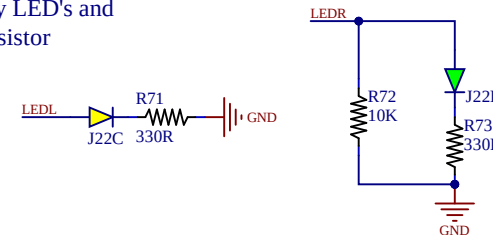
PHY decoupling capacitors



ESD protection



Activity LED's and strap resistor

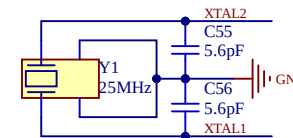


References

- Los resistores de pull up en los pines MODE0, MODE1 y MODE2 son resistores de strap que configuran al LAN8742 en el modo "all capable" con la autonegociación activada.
- LEDL cuenta con un resistor de strap que configura el pin REFCLKO como salida de reloj de referencia para la capa MAC, si se deja flotado o conectado a VDD2A, el pin indica interrupciones nINTSEL.
- Pull-Up interna en NRST.

See <https://www.st.com/en/evaluation-tools/nucleo-h563zi>
See https://www.microchip.com/downloads/DS_LAN8742_00001989A.pdf

External crystal



Title: Ethernet Interface (ETH).SchDoc

Project: MEC.PrjPcb

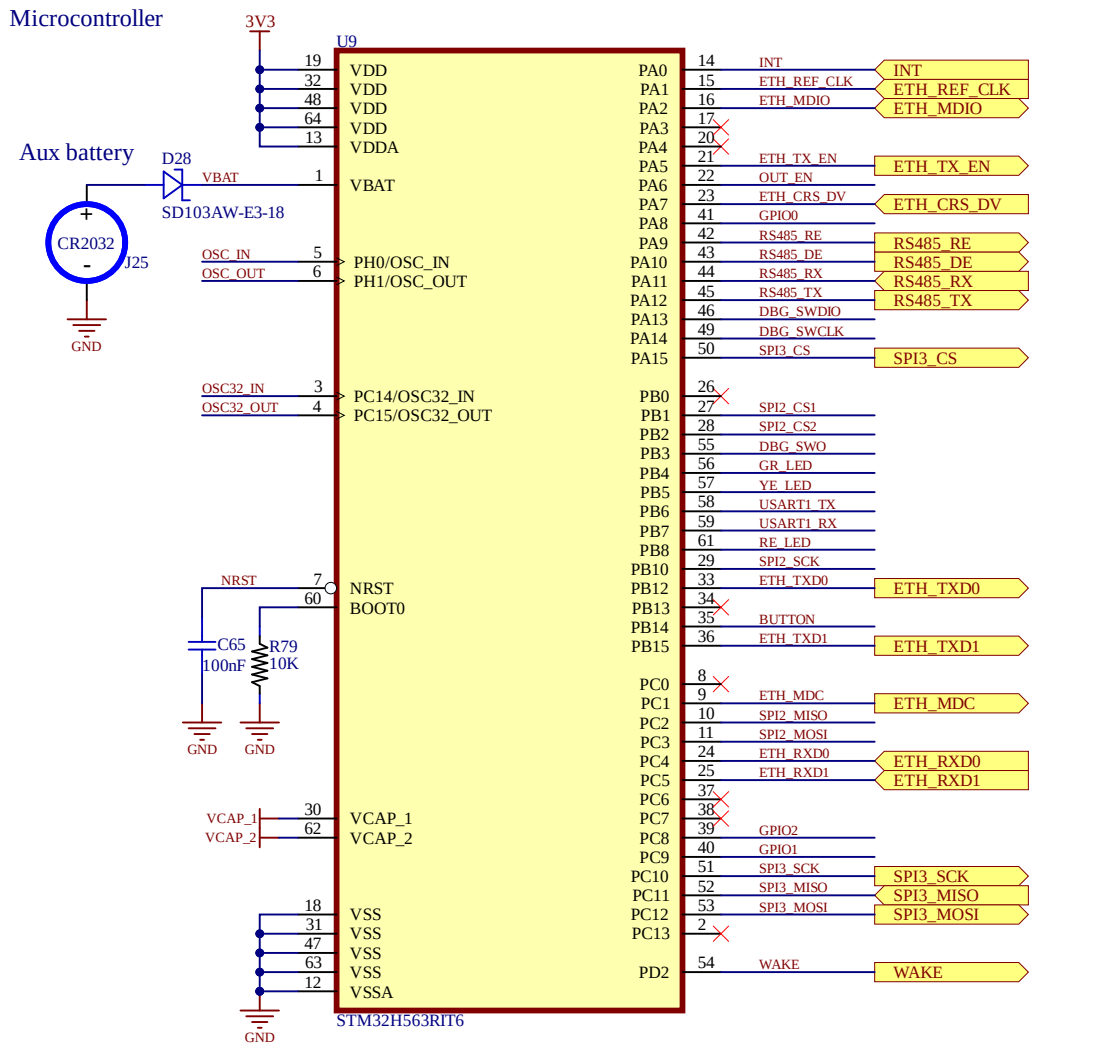
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Microcontroller

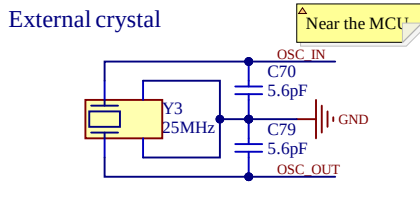


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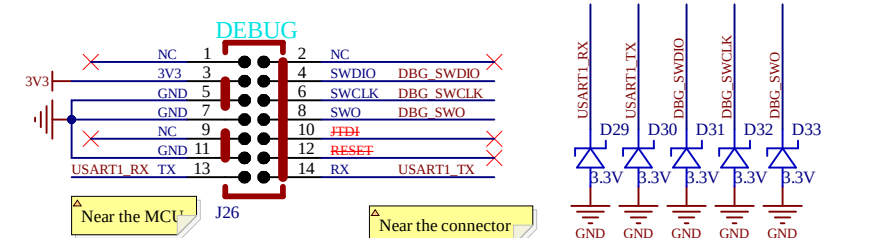
See <https://www.st.com/en/evaluation-tools/nucleo-h563zi>

Pull-Up interna en NRST.

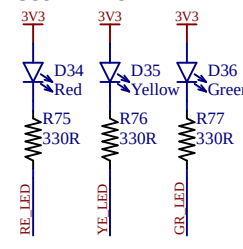
External crystal



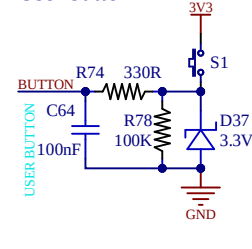
Serial wire interface



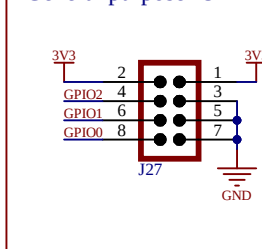
User LED's



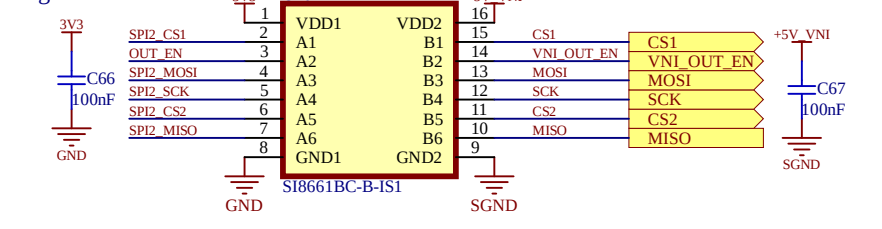
User button



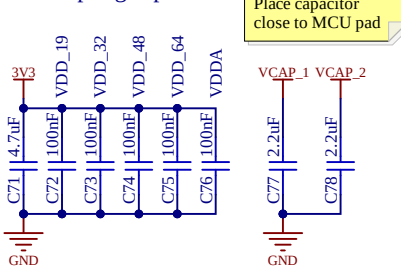
General purpose IO



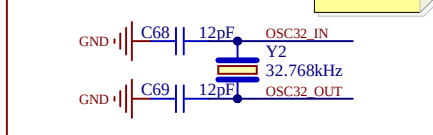
Digital isolator



Decoupling capacitors



External RTC



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